Dr. A. Arun
Professor and Head

Contact

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Academic Qualifications: M.Sc., M.Phil., Ph.D., PGDCA

Teaching and Research Experience: 20 Years

Additional Responsibilities

1. Head of the Department – Till date
2. Campus Director- Faculty of Science Campus- 25.1.2021 – Till date
3. Coordinator, Establishment of Bio-Plastic centre, Alagappa University, Karaikudi-12.5.2020 – Till date
4. Internal Expert - Institutional Biosafety Committee (IBSC), Alagappa University – 22.10.2020 to 22.10.2023.

5. Coordinator- Journal club Department of Microbiology, Alagappa University, Karaikudi.

6. Member – Coaching scheme for NET exam-, UGC schemes, Alagappa University, Karaikudi, Aug, 2016

7. Editor in Chief- ALU-Micro Fanzine, Departmental Magazine, Department of Microbiology, Alagappa University, Karaikudi.

8. Deputy Coordinator- Cultural club- Department of Microbiology, Alagappa University, Karaikudi.

9. Convener- Department Research Committee - Department of Microbiology, Alagappa University, Karaikudi.

10. Convener- Department purchase Committee - Department of Microbiology, Alagappa University, Karaikudi.

11. Convener- Department IQAC Committee - Department of Microbiology, Alagappa University, Karaikudi.

12. Convener- Department student affairs and counseling - Department of Microbiology, Alagappa University, Karaikudi.

13. Chairman of Valuation board for M.Sc., and M.Phil., Microbiology, Alagappa University

14. Deputy coordinator, Intellectual property cell, Alagappa University, 2018-19.

Areas of Research

- **Bioenergy**
- **Microbial fuel cell**
- **Bioremediation**
- **Biomass and Bioplastic**
- **Bioactive compounds**

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	Ph.D.	8	8
	M.Phil.	9	1
Project	PG	63	9
	UG / Others	-	-

Publications

International		National		Others
Journals	Conferences	Journals	Conferences	Books / Chapters / Monographs / Manuals
51	95	9	23	30

Cumulative Impact Factor	:	**270.845**
h-index	:	**20**
i10 index	:	**32**
Total Citations	:	**1166**

Funded Research Projects

Ongoing Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	DST	2022	2025	DST Bioenergy & H_2 MAP	1140.8705
2.	MHRD – SPARC (SPARC/2018-2019/P485/SL)	2019	2022	Cost Effective, Modified Microbial Bioplastics [Poly Hydroxyl Butyrate (PHB) And Poly Lactic Acid (PLA)] as an Alternative for the Petroleum Derived Plastics	56.57673

Ongoing DST-INSPIRE Fellowship under my Supervision

S. No	Agency and name of the candidate	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	DST-INSPIRE (IF190230) Ms. Abishpha Bora	2019	2024	Utilization Of Various Industrial Wastes For Different Bioenergy Production Using Aerobic Photohetero Micro Algal Isolates	**21.088**

Ongoing RUSA Post doc Fellowship under my Supervision

S. No	Agency and name of the candidate	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	RUSA 2.0 Dr. R. Karthik Raja	07.08.2019	Till date	Synthesizing of novel bioactive metallic nanoparticles using *Xenorhabdus* & *Photorhabdus* and evaluation of their therapeutic potential (antioxidant, antimicrobial and cytotoxic properties)	**14.50**

Consultancy Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	TNPCB	2008	2009	Microbial analysis of Vaigai river	1.0
2	Galaxy Research Technologies	15.05.2019	15.05.2020	Poly-hydroxyl-Alkonates (PHA) biopolymer research	3.6

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1.	RUSA Phase 2.0	2019	2021	Utilization of various lingo cellulosic wastes for the biodiesel production by microalgae under heterotrophic condition using Response surface methodology (RSM).	5,88,000
2.	UGC (MICR-2013-19708)	01.06.2015	01.06.2018	Biodegradable plastic (Poly – β – Hydroxybutyrate) production by marine microorganisms isolated from Tamilnadu coastal area	14.8

3.	DST-SERB (SB/YS/LS-47/2013)	Nov 2013	Nov 2016	Sequential Two-Stage Fermentation Process Producing Biohydrogen And Bioplastic (PHB) From Industrial Wastes	20.5
4.	AURF (AU:SO (P&D): Interdepartmental Research: 2017)	2017	2019	Utilization of *Pedalium murex* for Biodiesel and Silver Nanoparticles Production	3.0
5.	MBAI	2017	2018	Study on the enhancement of bioelectricity production by marine bacteria	0.2
6.	UGC	March 2004	March 2005	Production of Poly hydroxybutyrate by using various industrial wastes	0.8
7.	UGC	March 2009	March 2011	Polycyclic Aromatic Hydrocarbons (PAHs) Biodegradation by the Wild Basidiomycetes Fungi, Bacterial Isolates from Oil Spilled Soil, and their co-cultures: Comparative *In Vivo* and *In Silico* Approach	7.028
8.	TNSCST	2002		Utilization of different industrial wastes for the production of biopolymer (Polyhydroxy butrate) by using *Ralstonia eutropha*	0.05

9.	FLORA	2003		Microbial power generation by using marine sea mud – sea water interface and dairy industry waste – sea water interface	0.05
10	TNSCST	2004		Comparison and calibration of microbial power generation by utilizing biowaste	0.05
11	TNSCST	2005		Bio degradation of PAH by oil degrading bacteria	0.05
12	TNSCST	2007	2008	Comparative study on cyclosporine production by actinomycetes in liquid state and solid state conditions	0.05
13	TNSCST	2009	2010	A decolourization study of textile dye using ligninolytic enzyme of basidiomycetes fungi	0.05
14	TNSCST	2010	2011	Biohydrogen production by using keratin degraded effluent	0.05
15	TNSCST	2012	2013	Carbon sequestration (Flue gas utilisation) biohydrogen and biodiesel production using algal isolates	0.1
16	TNSCST	2017	2018	Assessment of Various pretreatment methods for enhanced biohydrogen production utilizing various industrial wastes	0.1
17	TNSCST	2018	2019	Biohydrogen and bioplastic (PHB) production from dairy industry wastwater by phototrophic bacteria	0.075

Distinctive Achievements / Awards

1. Young scientist award – DST-SERB-YSS-2013
2. Radio talk - About microbial Power generation in AIR- Madurai on 13.12.2004.

3. Travel grant (Rs 1, 17,383/- by DBT (DBT/CTEP/02/201600307 dated Apr 1, 016) to attend International Conference on Environmental science and Technology. Paper title: Anaerobic and subsequent photosynthetic process for biohydrogen and Bioplastic (PHB) production at Houston, US, American academy of sciences, Texas, Houston, USA; June 06-10, 2016.

4. Travel grant (Rs10,000/-) by Centre for International Co-operation in Science (CICS) – (DO\Lr.\TF-V\2015-16 dated 3.2.16) to attend 5th annual International Conference on sustainable energy and environmental sciences-2016 (SEES 2016) at Singapore organized by Global science and technology forum, Singapore (GSTF); 22– 23Feb, 2016.

5. Dr. APJ Abdul Kalam Award for scientific excellence -2018 by Marina Labs, Chennai, Tamilnadu, India on 13.10.18.

6. Travel grant (Rs1,00,000/-) by Alagappa University under RUSA 2.0 scheme to present a research paper at Sixth International Symposium Frontiers in Polymer Science organized by Elsevier Publications and Materials today at Budapest, Hungary on May 05-08th 2019.

7. Travel grant (Rs1,95,400/-) by Alagappa University under RUSA 2.0 scheme to for visiting the collaborating SPARC PI Institution and to give Invited lecture at Tianjin University, Tianjin, China from 25.10.2019 to 04.11.2019.

Events organized in leading roles

Number of Seminars / Conferences / Workshops / Events organized: 22

1. International – 07
2. National – 06
3. State level – 08
4. Regional/others – 01

Events Participated

Conferences / Seminars / Workshops: 58

Overseas Exposure / Visits

1. Visited Singapore to present a research paper in 5th annual International Conference on sustainable energy and environmental sciences-2016 (SEES 2016) at Singapore organized by Global science and technology forum, Singapore (GSTF); 22–23Feb, 2016.
2. Visited America to present a research paper in International Conference on Environmental science and Technology. Paper title: Anaerobic and subsequent photosynthetic process for biohydrogen and Bioplastic (PHB) production at Houston, US, American academy of sciences, Texas, Houston, USA; June 06-10, 2016.
3. Visited Hungary to present a research paper in Sixth International Symposium Frontiers in Polymer Science organized by Elsevier Publications and Materials today at Budapest, Hungary on May 05-08th 2019.
4. Visited China from 25.10.2019 to 04.11.2019 as SPARC PI and gave Invited lecture at Tianjin University, Tianjin, China (31.10.2019) and at Beijing Forestry University, Beijing, China (1.11.2019).

Membership in

Professional Bodies

1. Life Member: Association of Microbiologists of India (Life Membership No. 1354/2001).
2. Life Member: American Society for Microbiology (Membership ID: 56653553)
3. Life Member: South Indian Microbiologist Association (Life Membership No. SIMAMDU00040).

Editorial Board

- Guest Editor –Recent Advancements in Microbial Fuel Cells. 2022. Energies. (Impact Factor 3.004).
- Guest Editor - Recent Advancements in Microbial Fuel Cells - Energies / Vol. 28 issue 26, special issue "RTB2019". From Page 33866 to 33966, Springer Publications.2020 (Impact Factor 4.223).

- Guest Editor -3rd International Conference on Recent Trends in Microbiology (RTM-2019)- Burgeoning trends in Microbiology and Agriculture Sciences - Biocatalysis and Agricultural Biotechnology – Elsevier Publications.2020.

- Blue Biotechnology Journal - Nova Science Publishers, Inc. United States of America.
- Journal of Cell Science & Molecular Biology - Open Science Publications- India- 10.8.13.
- Editor, Alagappa University Journal of Biological Sciences (AUJBS), Alagappa University, Karaikudi.
- Editorial Board Member, Multidisciplinary journal, The Thassim Beevi Abdul Kader College for Women, Kilakarai- 20.2.17.
- Editorial Board Member, Biotechnology and Bioinformatics. Mediterranean Journals, 16192 Coastal Hwy, Lewes, DE 19958, United States – 09.08.2021.

Resource persons in various capacities

Number of Invited / Special Lectures delivered: 39

Others

1. Articles published in Newspapers / Magazines: 0
2. No. of PhD Thesis evaluated: 09
3. No. of PhD Public Viva Voce Examination conducted: 10

Sequence submitted in GenBank: 150

Bacteria – 43; Fungi – 104; Microalgae – 2; Protein – 1

Books Editing

1. Biodegradable Polymers, Blends and Biocomposites: Trends and Applications. CRC Press Taylor and Francis Group (USA) (9781032302492).
2. Enzymes incorporated nanoparticles their fundamentals concepts, synthesis & applications, Elsevier Academic Press (The Netherlands).
3. Algae Materials. Elsevier Academic Press (The Netherlands).

Recent Publications

1. Madhumitha Chidhambaram, Karmegam Natchimuthu, Biruntha Muniyandi, Arun Alagarsamy, Abdulaziz A Al. Kheraif, Woong Kim, Kumar Ponnuchamy. Extraction, identification, and environmental risk assessment of microplastics in commercial toothpaste. 2022. Chemosphere. 296: 133976. **(IF – 7.086).** https://doi.org/10.1016/j.chemosphere.2022.133976

2. Boobalan Thulasinathan, Tamilmani Jayabalan, Nagarajan Arumugam, Mohan Rasu Kulanthaisamy, Woong Kim, Ponnuchamy Kumar, Muthusamy Govarthanan, **Arun Alagarsamy**. Wastewater substrates in microbial fuel cell systems for carbon-neutral bioelectricity generation: An Overview. 2022. Fuel. Volume 317, 1 June 2022, 123369**. DOI:** https://doi.org/10.1016/j.fuel.2022.123369 **(IF 6.609).**

3. Maya M.R., V. Ananthi, **A. Arun**, P. Kumar, M. Govarthanan, K. Rameshkumar, V. Veeramanikandan , P. Balaji. 2022. Protective efficacy of *Capsicum frutescens* fruits in pancreatic, hepatic and renal cell injury and their attenuation of oxidative stress in diabetic rats. Journal of Taibah University for Science (TUSC). 15:1, 1232-1243, DOI: https://doi.org/10.1080/16583655.2021.2024998 **(IF 2.688).**

4. Murugan Sethupathi, Boobalan Thulasinathan, Nallathambi Sengottuvelan, Kumar Ponnuchamy, Franc Perdih, **Arun Alagarsamy**, and Muthusamy Karthikeyan. 2021. Macrocyclic "tet a"-Derived Cobalt(III) Complex with a N,N′-Disubstituted Hexadentate Ligand: Crystal Structure, Photonuclease Activity, and as a Photosensitizer. 2021. ACS Omega. DOI: https://doi.org/10.1021/acsomega.1c05306 **(IF 3.512).**

5. V. Thirumal, R. Yuvakkumar, P. Senthil Kumar, G. Ravi, **A. Arun**, Ramesh K. Guduru, Dhayalan Velauthapillai. 2022. Heterostructured two dimensional materials of MXene and Graphene by hydrothermal method for efficient hydrogen production and HER activities. International Journal of Hydrogen Energy. ISSN 0360-3199. **(IF 5.816). DOI:** https://doi.org/10.1016/j.ijhydene.2021.12.045

6. Maya. M.R., K. Rameshkumar , V. Veeramanikandan , T. Boobalan , M. Kumar , M. Eyini , **A. Arun**, Arivalagan Pugazhendhi, P. Balaji. 2022. Evaluation of Antioxidant, anti-inflammatory, and anti-hyperglycemic effects of *Wattakaka volubilis* Linn. f.

Process Biochemistry. 112. 183–191 **(IF 3.757). DOI:** https://doi.org/10.1016/j.procbio.2021.12.001

7. K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh, Kunyu Zhang, Sudhakar Muniyasamy, A. Pugazhendhi, J. Jeyakanthan, Kumar Ponnuchamy, M. Govarthanan, **A. Arun**. 2021. Production and Characterization of biodegradable Polyhydroxybutyrate by *Micrococcus luteus* isolated from marine environment. International Journal of Biological Macromolecules. **(IF – 6.953).** DOI: https://doi.org/10.1016/j.ijbiomac.2021.07.029

8. Abhispa Bora, T. Angelin Swetha, K. Mohanraasu, G. Sivaprakash, P. Balaji, A. Arun. Chapter 17. Microbial production of biohydrogen ($BioH_2$) from waste-activated sludge: Processes, challenges and future approaches. 2021. Advanced Functional Materials for Hydrogen Production, Conversion and Storage. Wiley publication. (Accepted).

9. Barik, A., Biswal, D., **Arun, A**. and Balasubramanian, V. (2021). Biodetoxification of Heavy Metals Using Biofilm Bacteria. In Environmental and Agricultural Microbiology (eds B.B. Mishra, S.K. Nayak, S. Mohapatra and D. Samantaray). Pages-39-61. 24 August 2021. ISBN:9781119526230. Scrivener Publishing LLC. DOI: https://doi.org/10.1002/9781119525899.ch3

10. R. Karthik Raja, Selcuk Hazir, R. Govindan, G. Balasubramani, **A. Arun**, (2022). Chapter 22: Green nanotechnology for the environment. In Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P- 461-478. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. DOI: https://doi.org/10.1016/B978-0-12-823426-6.00006-1

11. T. Angelin Swetha, K. Mohanrasu, Abhispa Bora, **A. Arun**. (2022). Chapter 20: Enzyme-incorporated nanotechnology in wastewater treatment. In Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P 415-438. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. DOI: https://doi.org/10.1016/B978-0-12-823426-6.00021-8

12. K. Mohanrasu, R. Guru Raj Rao, G. Sivaprakash, G.H. Dinesh , **A. Arun**. (2022). Chapter 16: Microbial bio-based polymer nanocomposites for food industry applications. In

Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P – 331-354. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. Doi: https://doi.org/10.1016/B978-0-12-823426-6.00012-7

13. G. Sivaprakash, K.Mohanrasu, G.H.Dinesh , **A. Arun**. (2022). Chapter 15: Microbial nanotechnology in food Industry: Antimicrobial packaging. In Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P 311-329. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. DOI: https://doi.org/10.1016/B978-0-12-823426-6.00002-4

14. Abhispa Bora, K.Mohanrasu, T. Angelin Swetha, **A. Arun**. (2022). Chapter 21: Microbes incorporated nanomaterials for water purification. In Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P- 439-459. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. DOI: https://doi.org/10.1016/B978-0-12-823426-6.00001-2

15. V. Ananthi, K.Mohanrasu, G.H. Dinesh, **A. Arun.** (2022). Chapter 17: Pathogen identification through surface marker recognition methods. In Handbook of Microbial Nanotechnology. (Ed. Chaudhery Hussain). P-415- 438. ISBN: 9780128234266, Academic Press, Publishing Date: March 2022. DOI: https://doi.org/10.1016/B978-0-12-823426-6.00014-0

16. Ramalingam Karthik Raja, Phuong Nguyen-Tri, Govindasamy Balasubramani, **Arun Alagarsamy**, Selcuk Hazir, Safa Ladhari, Alireza Saidi, Arivalagan Pugazhendhi, Arulandhu Anthoni Samy**. 2021.** SARS-CoV-2 and its new Variants: A Comprehensive Review on Nanotechnological Application Insights into Potential Approaches. Applied Nanoscience. **(IF – 3.674).** DOI: https://doi.org/10.1007/s13204-021-01900-w

17. Van-Huy Nguyen, P. Prema, T. Boobalan, A. Arun, K. Ramesh Kumar, R. Suresh Babu, V. Veeramanikandan, P. Balaji. 2022. Green tea mediated synthesis of gold nanoparticles with potent anti-proliferative effect against PC-3 human prostate cancer cells. Materials Letters. DOI: https://doi.org/10.1016/j.matlet.2021.130882 **(IF – 3.423).**

18. Arumugam Nagarajan, Boobalan Thulasinathan, Pugazhendhi Arivalagan, **Arun Alagarsamy**, Jothi Basu Muthuramalingam, Suganya Devi Thangarasu, Kavitha Thangavel. 2021. Particle size influence on the composition of sugars in corncob hemicellulose hydrolysate for xylose fermentation by *Meyerozyma caribbica*.

Bioresource Technology, Volume 340, 125677, ISSN 0960-8524, https://doi.org/10.1016/j.biortech.2021.125677 **(IF – 9.642).**

19. Premnath. N, K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh, G. Siva Prakash, V. Ananthi, Kumar Ponnuchamy, Govarthanan Muthusamy, **A. Arun**. (2021). A Crucial Review on Polycyclic Aromatic Hydrocarbons - Environmental Occurrence and Strategies for Microbial Degradation. Chemosphere. DOI: 10.1016/j.chemosphere.2021.130608 **(IF – 7.086).**
20. V. Ananthi, P. Balaji, Raveendran Sindhu, Sang-Hyoun Kim, Arivalagan Pugazhendhi , **A. Arun**. (2021). A critical review on different harvesting techniques for algal based biodiesel production. Science of the Total Environment. 78; 146467 DOI: https://doi.org/10.1016/j.scitotenv.2021.146467 **(IF – 7.963).**
21. Saravanan Soorangkattan, Kamala Dhasan Nalluchamy, Arumugam Nagarajan, Boobalan Thulasinathan,Maruthupandian Jayabalan, Jothi Basu Muthuramalingam, **Arun Alagarsamy**, Muthuchelian Krishnasamy. (2021). In situ conservation of endangered tree species (*Elaeocarpus venustus* Bedd.) habitated in Agasthiyamalai Biosphere Reserve, Southern Western Ghats, India. Environmental Science and Pollution Research. DOI: https://doi.org/10.1007/s11356-021-13227-8 **(IF – 4.223)**.
22. Saravanan Soorangkattan, Kamala Dhasan Nalluchamy,Sudha Arumugam, Chandrasekaran Sivagnanam, Boobalan Thulasinathan, Satheesh Murugan Ramu, **Arun Alagarsamy**, Jothi Basu Muthuramalingam.(2021). Studies on the influence of natural resource utilization by humans on foraging behavior of honey bees at rural ecosystems. Environmental Science and Pollution Research. https://doi.org/10.1007/s11356-021-13192-2 **(IF – 4.223)**.
23. Boobalan Thulasinathan, Tamilmani Jayabalan, Murugan Sethupathi, Woong Kim, Sudhakar Muniyasamye, Nallathambi Sengottuvelan, Samsudeen Nainamohamed, Kumar Ponnuchamy, **Arun Alagarsamy**. (2021). Bioelectricity Generation by Natural Microflora of Septic Tank Wastewater (STWW) and Biodegradation of Persistent Petrogenic Pollutants by Basidiomycetes Fungi: An Integrated Microbial Fuel Cell System. Journal of Hazardous Materials. Volume 412, 25228, ISSN 0304-3894**.** https://doi.org/10.1016/j.jhazmat.2021.125228 **(IF – 10.588)**.

24. Mohanrasu K., Rao R.G.R., Sudhakar M., Raja R., Jeyakanthan J., **Arun A.** (2020) Marine Microbial Pharmacognosy: Prospects and Perspectives. In: Nathani N.M., Mootapally C., Gadhvi I.R., Maitreya B., Joshi C.G. (eds) Marine Niche: Applications in Pharmaceutical Sciences. Springer, Singapore. https://doi.org/10.1007/978-981-15-5017-1_5 . ISBN 978-981-15-5016-4

25. Premnath. N, K. Mohanrasu, R. Guru Raj Rao, G.H. Dinesh , G. Siva Prakash, Arivalagan Pugazhendhi, J. Jeyakanthan, Muthusamy Govarthanan, Ponnuchamy Kumar, **A. Arun**. 2020. Effect of C/N Substrates for enhanced Extracellular Polymeric Substances (EPS) Production and Poly Cyclic Aromatic Hydrocarbons (PAHs) degradation. Environmental Pollution. 116035, ISSN 0269-7491 **(IF – 8.071). DOI:** https://doi.org/10.1016/j.envpol.2020.116035

26. Sabarathinam Shanmugam, Thangavel Mathimani, Eldon R. Rene, V Edwin Geo, **A. Arun**, Kathirvel Brindhadevi. 2021. Biohythane production from organic waste: Recent advancements, technical bottlenecks and prospects. International Journal of Hydrogen Energy. ISSN 0360-3199 (**IF: 5.816**). https://doi.org/10.1016/j.ijhydene.2020.10.132

27. Ramalingam Karthik Raja, **Alagarsamy Arun**, Mustapha Touray, Sebnem Hazal Gulsen, Harun Cimen, Baris Gulcu, Canan Hazir, Dilipkumar Aiswarya, Derya Ulug, Ibrahim Cakmak, Harry K. Kaya, Selcuk Hazir. Antagonists and defense mechanisms of entomopathogenic nematodes and their mutualistic bacteria. Biological Control. Volume 152, 2021, 104452, ISSN 1049-9644, https://doi.org/10.1016/j.biocontrol.2020.104452 . **(IF – 3.687).**

28. V. Ananthi , Arivalagan Pugazhendhi , **A. Arun**. Efficacy of chemical factors on production and extraction of biodiesel by Microalgae. 2020. International Journal of Energy Research. **(IF – 5.164).** DOI: https://doi.org/10.1002/er.6097

29. K. Mohanrasu, R. Guru Raj Rao, Rathinam Raja, and **A. Arun**. Bioremediation Process by Marine Microorganisms. In book: Encyclopedia of Marine Biotechnology. Se-Kwon Kim (Ed.). 2020. Wiley-Blackwell. ISBN: 978-1-119-14377-2. https://doi.org/10.1002/9781119143802.ch100

30. M. Sivagami, M. Selvambigai, U. Devan, A. Antony Joseph Velangani , N. Karmegam , M. Biruntha , **A. Arun,** W. Kim, M. Govarthanan, P. Kumar. 2021. Extraction of microplastics

from commonly used sea salts in India and their toxicological evaluation. https://doi.org/10.1016/j.chemosphere.2020.128181 . Chemosphere. **(IF – 7.086).**

31. Samsudeen Naina Mohamed, Dinesh Bejjanki, Muthukumar K, T.K. Radhakrishnan T.K. Radhakrishnan, **Arun Alagarsamy**, Arivalagan Pugazhendhi. 2020. Simultaneous Bioelectricity Generation and Water Desalination Using *Oscillatoria* Sp. In Photosynthetic Microbial Desalination Cell. Science of the Total Environment. https://doi.org/10.1016/j.scitotenv.2020.142215 **(IF – 7.963).**

32. Khadeeja Parveen kallarakkal, muthukumar k, Arivalagan Pugazhendhi, **Arun Alagarsamy**, Samsudeen Naina Mohamed. 2020. Enhancement of biobutanol production using mixotrophic culture of *Oscillatoria* sp. in cheese whey water. Fuel. https://doi.org/10.1016/j.fuel.2020.119008 **(IF – 6.609)**.

33. V Ananthi, Kathirvel Brindhadevi, Arivalagan Pugazhendhi, **Arun Alagarsamy**. 2020. Impact of abiotic factors on biodiesel production by microalgae. Fuel. https://doi.org/10.1016/j.fuel.2020.118962 **(IF – 6.609)**.

34. Ramu Satheesh Murugan, Gujuluva Hari Dinesh, Thulasinathan Boobalan, Thondi Rajan Angelin Swetha, Ponnuchamy Kumar, Arivalagan Pugazhendhi, **Alagarsamy Arun**. 2020. Dark fermentative biohydrogen production from rice mill wastewater. International Journal of Energy Research. http://dx.doi.org/10.1002/er.5829 **(IF – 5.164)**.

35. V Ananthi, Rathinam Raja, Isabel S Carvalho, Kathirvel Brindhadevi, Arivalagan Pugazhendhi; **Arun Alagarsamy**, 2020. A realistic scenario on microalgae-based biodiesel production: Third generation biofuel. Fuel. https://doi.org/10.1016/j.fuel.2020.118965 **(IF – 6.609)**.

36. Rathinam Raja , Shanmugam Hemaiswarya , Sekaran Sridhar , **Arun Alagarsamy** , Venkatesan Ganesan , Sanniyasi Elumalai and Isabel S Carvalho. 2020. Evaluation of Proximate Composition, Antioxidant Properties, and Phylogenetic Analysis of Two Edible Seaweeds, Smart Science. Taylor and Francis Publication. https://doi.org/10.1080/23080477.2020.1795338

37. Ramu Satheesh Murugan, Gujuluva Hari Dinesh, Ramalingam Karthik Raja, Ebenezer Samuel James Obeth, Abhispa Bora, Naina Mohammed Samsudeen , Arivalagan Pugazhendhi, **Alagarsamy Arun**. 2020. Dark fermentative biohydrogen production by *Acinetobacter junii*-AH4 utilizing various industry wastewaters. International Journal of Hydrogen Energy. https://doi.org/10.1016/j.ijhydene.2020.07.073 (**IF: 5.816**).

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